Solution Manual of Probability

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Chapter 2 in [1]

Exercise (2.1-7)**.** Let a random experiment be the casting of a pair of unbiased six-sided dice, and let X equal the smaller of the outcomes if they are different and the common value if they are equal.

1. With reasonable assumptions, find the p.m.f. of X.
2. Let Y equal the range of the two outcomes (i.e. the absolute value of the difference of the largest and the smallest outcomes). Determine the p.m.f. $g(y)$ of Y for $y = 0, 1, 2, 3, 4, 5$.

Solution. 題目表示出 X 是六面骰骰兩次中最小點數的隨機變數。因為合計有 36 種可能狀況，其中可以知道數字為 1 的情況有 11 種，以此類推。更具體一點為，考慮 $X = x$，那麼我們可以將

$$P(X \geq x) - P(X \geq x + 1) = P(X = x)$$

用來計算，因此

$$(7 - x)^2 - (6 - x)^2 = 13 - 2x$$

所以 X 的 p.m.f. 為

$$\frac{13 - 2x}{36}$$

而 Y 的 p.m.f. 則可以直接計數 (暫時沒想到其他辦法) 得到

$$P(Y = y) = \begin{cases} \frac{1}{6}, & y = 0, \\ \frac{6-y}{18}, & y = 1, 2, 3, 4, 5. \end{cases}$$

□

Exercise (2.1-11)**.** In a lot (collection) of 100 light bulbs, there are 5 bad bulbs. An inspector inspects 10 bulbs selected at random. Find the probability of finding at least one defective bulb. Hint: First compute the probability of finding no defectives in the sample.

Solution. 抽 10 個至少壞 1 個的機率

$$P = 1 - \text{全部都是好的機率} = 1 - P'$$

全部是好的機率就很好算了。

$$P' = \frac{95}{100} \times \frac{94}{99} \cdots \times \frac{86}{91}$$

藉由計算機，

$$P = 0.4163$$

□

Exercise (2.1-13)**.** Let X be the no. of accidents per week in a factory. Let the p.m.f. of X be

$$f(x) = \frac{1}{(x+1)(x+2)}, \quad x = 0, 1, 2, \cdots .$$

Find the conditional probability of $X \geq 4$, given that $X \geq 1$. Hint:Write $f(x) = 1/(x+1) - 1/(x+2)$.

Solution.

$$P(X \geq 4 | X \geq 1) = \frac{P(X \geq 4 \cap X \geq 1)}{P(X \geq 1)} = \frac{P(X \geq 4)}{P(X \geq 1)}$$

那麼先算 $X \geq 4$，

$$P(X \geq 4) = \sum_{4}^{\infty} \frac{1}{n+1} - \frac{1}{n+2} = \frac{1}{5}$$

以及

$$P(X \geq 1) = \sum_{1}^{\infty} \frac{1}{n+1} - \frac{1}{n+2} = \frac{1}{5} = \frac{1}{2}$$

所以答案為 $\frac{2}{5}$。 □

Exercise (2.1-15)**.** A professor gave her students six essay questions, from which she will select three for a test. A students has time to study for only three of these questions. What is the probability that, of the questions studied,

1. at least one is selected for the test?
2. all three are selected for the test?
3. exactly two are selected for the test?

Solution. 教授挑選題目的種類為 $C_3^6 = 20$。

1. 機率為 1 減掉全部都沒中的機率。全部都沒中的機率為

   $$\frac{20}{20 \times 20} = \frac{1}{20}$$

   所以至少有一個被選中的機率為

   $$1 - \frac{1}{20} = \frac{19}{20}$$

2. 全部被選中的機率為

   $$\frac{20}{20 \times 20} = \frac{1}{20}$$

3. 恰好兩個被選中的機率為

   $$\frac{C_2^6 \times C_1^4 \times C_1^3}{20 \times 20} = \frac{180}{400} = \frac{9}{20}$$

□

References

[1] R. V. Hogg and E. A. Tanis, *Probability and Statistical Inference*, Pearson, 2010.